

Incident definitions and thresholds

Generated from the committed trips.geojson (192,681 features; 19 crash points; 3,734 flagged braking segments).

Source: trips.geojson
Scope: Existing classified incidents and stored trip measurements

1. What this report is based on

This report does not rerun incident detection. Crash points are taken from GeoJSON features with **event_type = crash**. Sudden-braking rows are taken from segment features with **is_braking = true**. The report only derives contextual fields such as surrounding speed from timestamps and stored segment values.

2. Crash / fall definition

Rule / field	Current definition	Why it matters
Impact threshold	acc_y ≥ 6.0 g	Raw accelerometer impact spike enters crash-candidate clustering.
Impact clustering	Consecutive threshold samples grouped when gap ≤ 1.0 s	Prevents one physical impact from becoming many candidate spikes.
Burst grouping	Separate throws grouped when gap ≤ 3.0 s	Allows rapid multiple impacts to be evaluated as one burst.
Accelerometer settling	Post-impact window up to 3.0 s; stable for about 1.0 s; range ≤ 3.0 g	Requires the accelerometer signal to settle after the impact.
GPS stop	GPS speed ≤ 5.5 km/h for ≥ 1.0 s and coordinate spread ≤ 5.0 m	Separates a fall/stop from a jolt while continuing to ride.
Motion before impact	Moving events use wheel rotation before the burst; genuinely stationary-before events are allowed when pre-impact GPS also confirms the bike was stationary.	Allows stationary falls/tips while retaining a motion check for moving falls.
Stationary-before Tip	If the bike is stationary before impact and remains stationary after the impact (with accelerometer settling and GPS stop also confirmed), classify as Tip.	Separates a tip/fall of a stationary bike from moving falls.
Wheel stall / recovery	Moving events require a wheel stall gap ≥ 1.0 s and recovery ≤ 120.0 s. Tips do not require a recovery-time value.	Confirms that a moving fall is followed by a stop; stationary tips are handled by the dedicated Tip rule.
Impact-speed estimate	Wheel-rotation estimate; events at/above 40 km/h are not trusted	This is not the GNSS speed field.

3. Crash intensity / severity

Peak [g]	Label
6.0–<8.0 g	Minor
8.0–<11.0 g	Hard
≥11.0 g	Severe

4. Crash classification

Condition	Classification
crash_type = "Tip"	Tip
No crash_type; speed ≤1 km/h	Tip
No crash_type; >1–10 km/h	Low-Speed Fall
No crash_type; >10 km/h	High-Speed Fall
No crash_type and no usable speed	Unclassified

A Tip is explicitly marked with crash_type = Tip when the bike was stationary before the impact and remains stationary afterward. Moving crash events currently have no crash_type value, so this report uses the available pre-impact or impact speed to distinguish Low-Speed Fall from High-Speed Fall. If no usable speed is available, the event is reported as Unclassified.

5. Crash outcome

outcome	Definition
Resolved	came_to_stop = true
Unresolved	unresolved = true
Unclassified	Neither outcome field establishes a state

The current generator exports a crash only after the required stop/recovery conditions are satisfied, so crash points produced by the current pipeline are expected to be Resolved. Unresolved is retained as a valid outcome for any record that explicitly contains unresolved = true.

6. Sudden braking definition

Rule / field	Current definition
Speed source	GNSS speed, smoothed with a 5-point rolling average
Braking intensity	(previous stored speed – current stored speed) / elapsed seconds; positive values only

Rule / field	Current definition
Minimum segment time	0.5 s
GNSS braking threshold	≥ 2.0 km/h/s
Intensity cap	50 km/h/s
Implausible speed jump	> 20 km/h between consecutive stored speeds resets the braking series

The report uses the existing `is_braking` flag. It does not apply the separate 5 km/h/s hotspot threshold from the hotspot-generation workflow, because that threshold is for hotspot inclusion rather than the raw braking flag.

7. Derived report fields

Field	How this PDF obtains it
Surrounding speed	Mean of stored Speed values from 5s before to 1s after the incident timestamp (not a symmetric window — see <code>crash_row</code>).
Avg / peak pre-impact deceleration	Mean / maximum of stored <code>braking_intensity</code> values in the 5s before a crash. These are GNSS braking values, not accelerometer impact force.
Full crash timestamp	Matched from <code>crash_time_str</code> to the nearest segment timestamp on the same trip when a full timestamp is not already stored on the crash point.
Braking coordinates	First coordinate of the flagged <code>LineString</code> segment; no new spatial detection is performed.

8. Important limitations

- The crash point location is explicitly marked `location_approximate = true` in the exported data.
- Latitude/longitude are stored to 6 decimal places, but that is numerical storage precision, not GPS accuracy. The source code comments put the practical GPS accuracy floor at roughly 1–3 m.
- `speed_at_impact_kmh` is a wheel-rotation estimate rather than the smoothed GNSS Speed field.
- The GeoJSON does not contain a crash-event deceleration measurement. The crash-table deceleration columns are therefore explicitly pre-impact GNSS braking metrics.
- Vehicle type is not part of the current committed GeoJSON segment/crash schema, so it is not fabricated in these reports.
- Stationary-before events are now eligible for Tip classification when pre-impact GPS also confirms the bike was stationary. Moving events retain the wheel-stall/recovery checks. The current generator only emits confirmed crash points; candidates that fail the required post-impact checks are not exported.

9. Source files

- Executable definitions: `generate_trips_geojson.py`.
- Frontend classification/outcome logic: `app.js`.
- Data: `trips.geojson`.